Distributed Systems

Syllabus

Lecture 01

Overview of Distributed Systems:
- Concepts of distributed systems
- Characteristics of distributed systems
- Classification of distributed systems
- Design goals and issues
- Architectures and models

Lecture 02

Communication and Naming in Distributed Systems
- Communication issues in distributed systems
- Naming issues in distributed systems

Lecture 03

Time and Physical Clocks; Clock Synchronization Algorithms
- Time and physical clocks
- Physical clock algorithms
- Logical clocks and Lamport clocks
- Vector clocks
- Clock usage

Lecture 04

Mutual Exclusion and Election Algorithms; Processes in Distributed Systems
- Mutual exclusion issues
- Election algorithms
- Process-related issues in distributed systems

Lecture 05

Transaction management and concurrency control; Recovery and fault tolerance in distributed systems
- Transactions in distributed systems
- Concurrency control issues in distributed systems
- Fault tolerance in distributed systems

Lecture 06

Recovery and fault tolerance in distributed systems; Consistency and replication
- Resilience of distributed systems
- Data replication needs and replica management issues
- Consistency models
- Consistency protocols

Lecture 07

Consistency and replication
- Consistency models

Lecture 08

Consistency and replication
- Consistency protocols

Lecture 09

Security in distributed systems
- Understanding security issues in distributed systems

Lecture 10

Security in distributed systems
- Security mechanisms for distributed systems

Lecture 11

RPC, CORBA, and RMI
- CORBA and RMI architectures
- CORBA and RMI programming models

Lecture 12

Web Services, SOA
- Web Services technology
- Introduction to SOA architecture
- Services
- Loose coupling model
- Service lifecycle
- Service classification
- Enterprise Service Bus (ESB)
- SOA-based architectural models
- Message exchange patterns in SOA

Textbook and references

[1] A. S. Tanenbaum, M. V. Steen, "Distributed Systems: Principles and Paradigms ", 2nd Edition, Prentice-Hall, 2007.
[2] G. Coulouris, J. Dollimore, T. Kinberg, G. Blair, "Distributed systems : Concept and Design", 5th Edition, Addison-Wesley, 2012.
[3] N.M. Josuttis, “SOA in Practice – The Art of Distributed System Design”, O’Reilly, 2007.